A PROJECT REPORT ON

"AI based internship recommendation Engine"

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ABSTRACT

The AI-Based Internship Recommendation Engine is a backend-driven intelligent system designed to provide personalized internship suggestions to students by analyzing their skills, preferences, and academic interests. The system is implemented using Java and Spring Boot**,** following **a** microservices-based architecture, where separate services handle student data and internship data.

The recommendation logic is implemented in the backend using a rule-based scoring algorithm**,** which evaluates the compatibility between student profiles and internship requirements. The system communicates with external services using REST APIs (via RestTemplate) to fetch student and internship data dynamically.

Each internship is assigned a match score based on multiple parameters such as skill matching, preferred location, domain, and internship type. The results are then sorted and returned in descending order of relevance.

The system is tested using Postman**,** ensuring API reliability and correctness. This project demonstrates how backend systems can incorporate intelligent recommendation techniques without requiring complex machine learning models, making it efficient, scalable, and easy to extend.

INTRODUCTION

In the modern job market, students often struggle to find internships that align with their skill sets and career aspirations. Most existing platforms require manual searching, filtering, and comparison, which is both time-consuming and inefficient.

To address this issue, the AI-Based Internship Recommendation Engine automates the process of identifying suitable internship opportunities. The system intelligently analyzes student profiles and compares them with internship requirements to generate personalized recommendations.

This project focuses on backend system design using Spring Boot, where multiple services interact with each other using REST APIs. The Recommendation Service acts as the core component, fetching data from:

- Student Service (student profile)
- Internship Service (available internships)

The system then applies a scoring algorithm to rank internships based on relevance. This approach not only reduces manual effort but also improves accuracy and efficiency in internship selection.

PROBLEM STATEMENT

Students face significant challenges in identifying internships that match their skills, preferences, and career goals. Traditional job portals provide large volumes of listings but lack personalized filtering, forcing users to manually analyze each opportunity.

There is a need for a system that:

- Automatically analyzes student profiles
- Matches them with relevant internships
- Provides ranked recommendations
- Reduces time and effort in searching

OBJECTIVE

- To design and implement a microservices-based backend system using Java and Spring Boot.
- To fetch and integrate data from multiple services (Student Service and Internship Service) using REST APIs (RestTemplate).
- To rank internships by relevance using a scoring mechanism and return sorted recommendations.
- To improve the efficiency of internship searching by automating the matching process.
- To ensure the system is scalable and easily integrable with frontend technologies like Angular.
- To test and validate APIs using Postman for accurate and reliable results.

SOFTWARE REQUIREMENTS

Development Environment

- Programming Language: Java
- Framework**:** Spring Boot
- Build Tool**:** Apache Maven

Architecture

- Microservices Architecture
- Layered Structure (Controller, Service, Model, DTO)

Backend Tools

- REST API Communication: Rest Template

Database (Optional / Based on your setup)

- MySQL

Testing Tools

- Postman

THEORY

The AI-Based Internship Recommendation Engine is based on the fundamental concept of recommendation systems, which are a key application area of artificial intelligence. A recommendation system is designed to analyze user data and suggest relevant items by identifying patterns, preferences, and similarities. In this project, the recommendation mechanism is implemented using a rule-based scoring approach, where internships are suggested based on how well they match a student's profile.

The system follows a microservices architecture, where different services operate independently and communicate through REST APIs. The Student Service provides information such as skills, preferred location, domain, and internship type, while the Internship Service provides details about available internships, including required skills and other attributes. The Recommendation Service acts as the core component, which integrates data from these services using RestTemplate, a Spring Boot utility for making HTTP requests. This separation of services improves scalability, modularity, and maintainability of the system.

The recommendation process begins by retrieving student data and internship data from their respective services. The system then preprocesses the data by converting skill strings into sets. This is done to ensure efficient comparison and eliminate duplicates. Converting skills into lowercase and trimming spaces ensures uniformity and avoids mismatches due to case sensitivity or formatting issues.

The core logic of the system lies in the scoring algorithm, which evaluates each internship based on multiple parameters. For every internship, the system compares the student's skills with the required skills of the internship. Each matching skill contributes a fixed score (for example, 10 points), making skill matching the most significant factor in the recommendation process. Additional parameters such as preferred location, domain, and internship type are also considered. If these attributes match between the student and the internship, additional points are added to the total score. This multi-criteria evaluation ensures that recommendations are not only based on technical skills but also on user preferences.

After calculating the score for each internship, the system stores the results along with their respective match scores. These results are then sorted in descending order using Java Stream API, ensuring that the most relevant internships appear at the top. The final output is a ranked list of internships, each accompanied by a match score, which helps users understand the relevance of each recommendation.

From a technical perspective, the system uses efficient data structures such as sets for skill matching and lists for storing results, along with functional programming features like streams for sorting and processing. This improves performance and readability of the code. The use of DTOs (Data Transfer Objects) ensures structured data exchange between services, while the layered architecture (Controller, Service, Model) promotes clean code organization.

Although the system is termed "AI-based," it primarily uses a rule-based intelligent approach rather than machine learning models. This makes the system lightweight, easy to implement, and suitable for backend-focused applications. However, the design allows future enhancements where machine learning techniques such as collaborative filtering or content-based learning can be integrated to improve recommendation accuracy.

Microservices Architecture

The system follows a microservices-based approach, where each component is independent:

1. Student Service

- Provides student details via API
- Example: /students/{id}

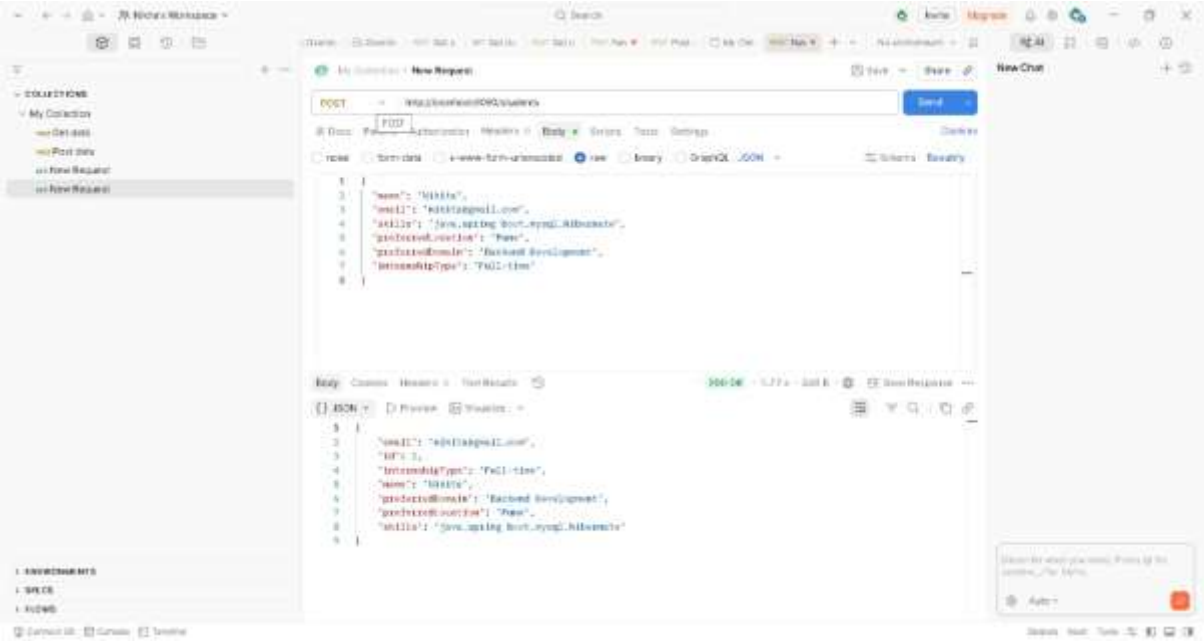
2. Internship Service

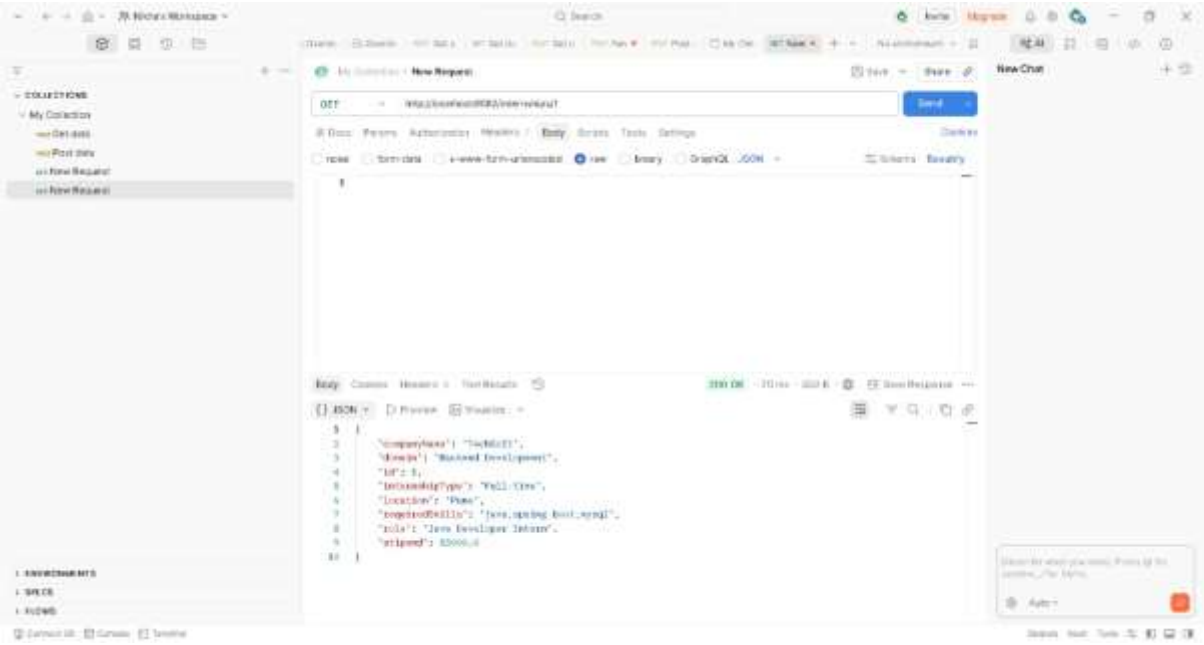
- Provides list of internships
- Example: /internships

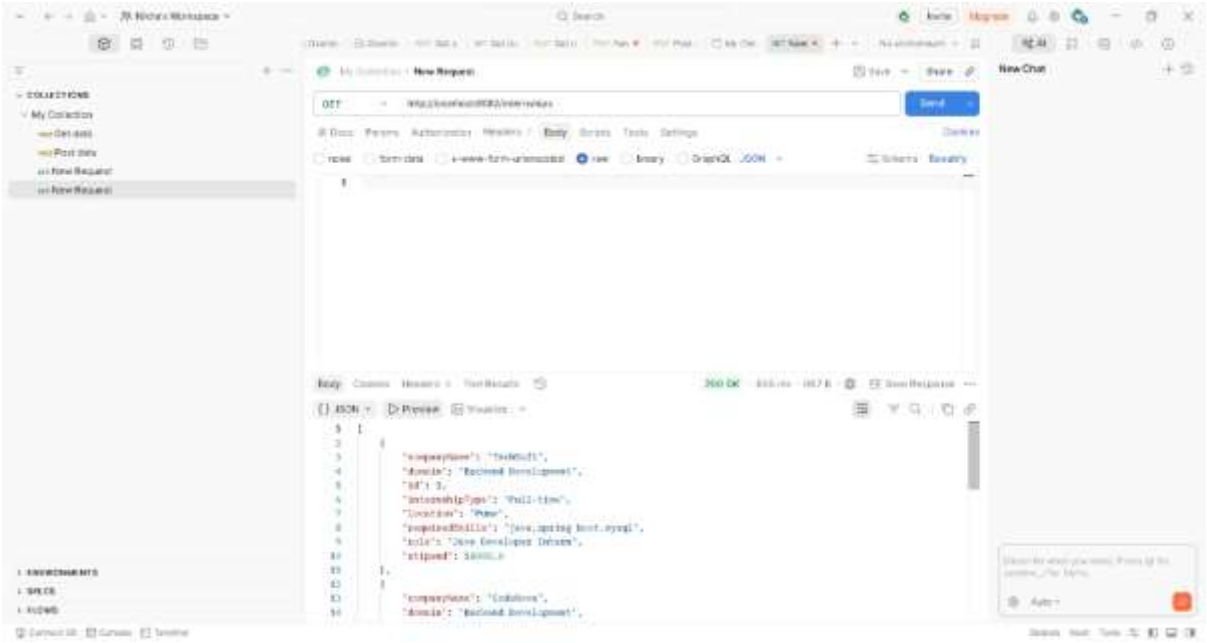
3. Recommendation Service

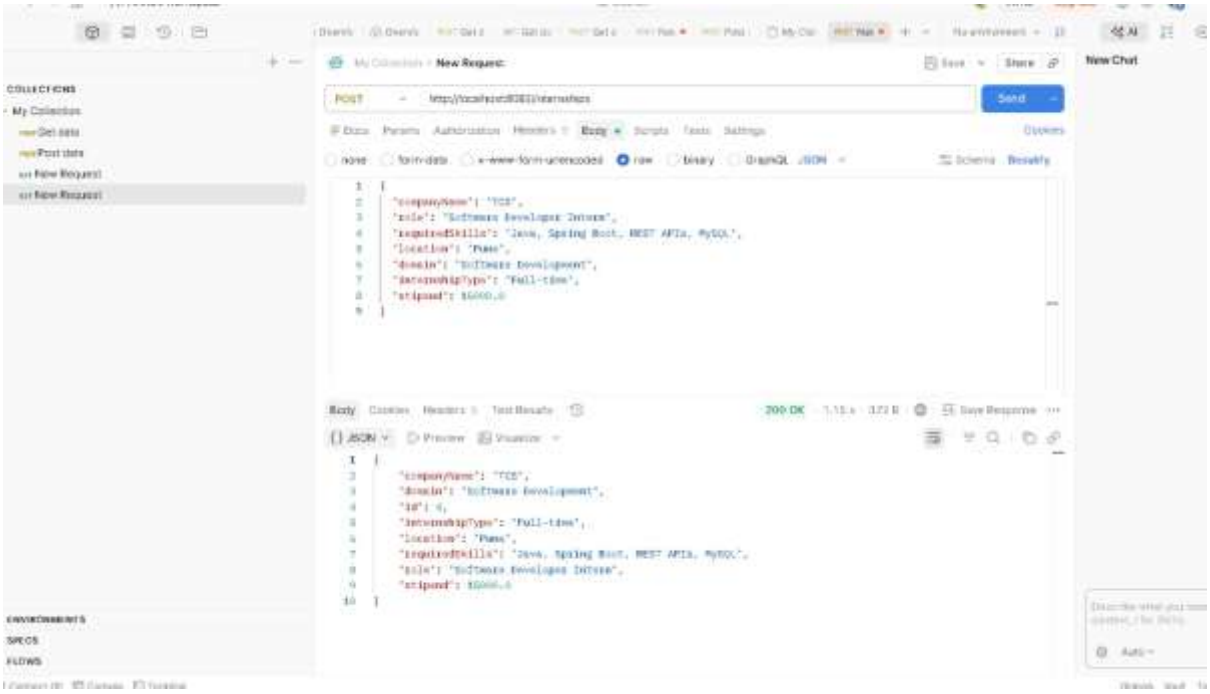
- Core logic implementation
- Fetches data from both services
- Generates recommendations

OUTPUT





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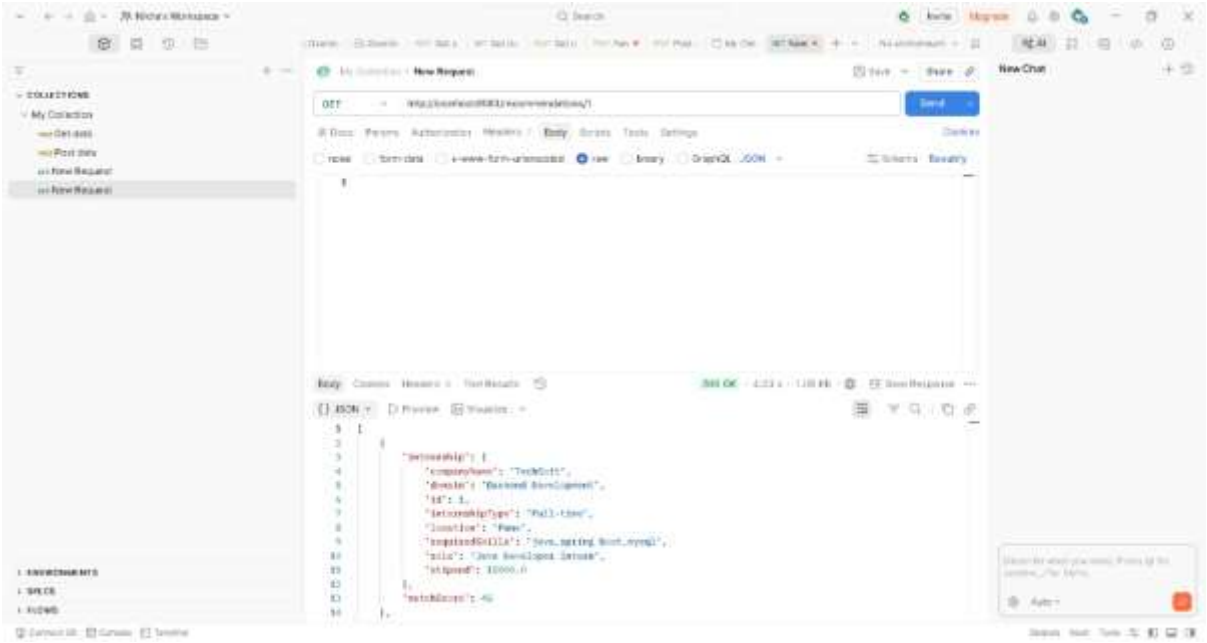
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New Chat

CONCLUSION

The AI-Based Internship Recommendation Engine successfully demonstrates how intelligent backend systems can be used to automate and improve the internship selection process for students. By leveraging a rule-based scoring algorithm, the system efficiently analyzes student profiles and matches them with relevant internship opportunities based on skills, preferences, and other criteria.

The use of microservices architecture and REST API communication ensures modularity, scalability, and ease of integration. The implementation using Java, Spring Boot, and Maven with a layered structure (Controller, Service, Model) provides a clean and maintainable codebase. Additionally, the use of Postman for API testing ensures that the system performs reliably and produces accurate results.

The project highlights that even without complex machine learning models, an effective recommendation system can be built using well-designed logic and structured data processing. It significantly reduces manual effort, saves time, and enhances decision-making for students searching for internships.